

# FREQUENCY

## VERIFIABLE PROOF-OF-WORK ENGINEERING DISCOVERY & HIRING

**Proof you can inspect. Skills you can trust.**

Team HM26-C3B5 · Members: [add names] · Colleges: [add colleges]

Selected problem · The proof gap in cybersecurity hiring

# THE PROOF GAP

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**Resumes and certificates show claims. They rarely show reproducible work, sound reasoning, or how a candidate explains a finding.**

## **A Mysuru hiring scenario**

A local team screens many applicants. A capable candidate without a familiar credential is easy to miss; an inflated claim is hard to verify.

Candidates lose visibility. Reviewers spend time rechecking. Recruiters lack comparable evidence.

# WHO IT SERVES

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## CANDIDATE

Demonstrates work through a finding, assessment and technical defense. Needs a clear path and low-bandwidth access.

## REVIEWER

Checks reproduction evidence, records a decision and scores defense. Human judgment remains authoritative.

## RECRUITER

Filters evidence-backed profiles and reviews sanitized proof before inviting candidates.

# FROM WORK TO PROOF

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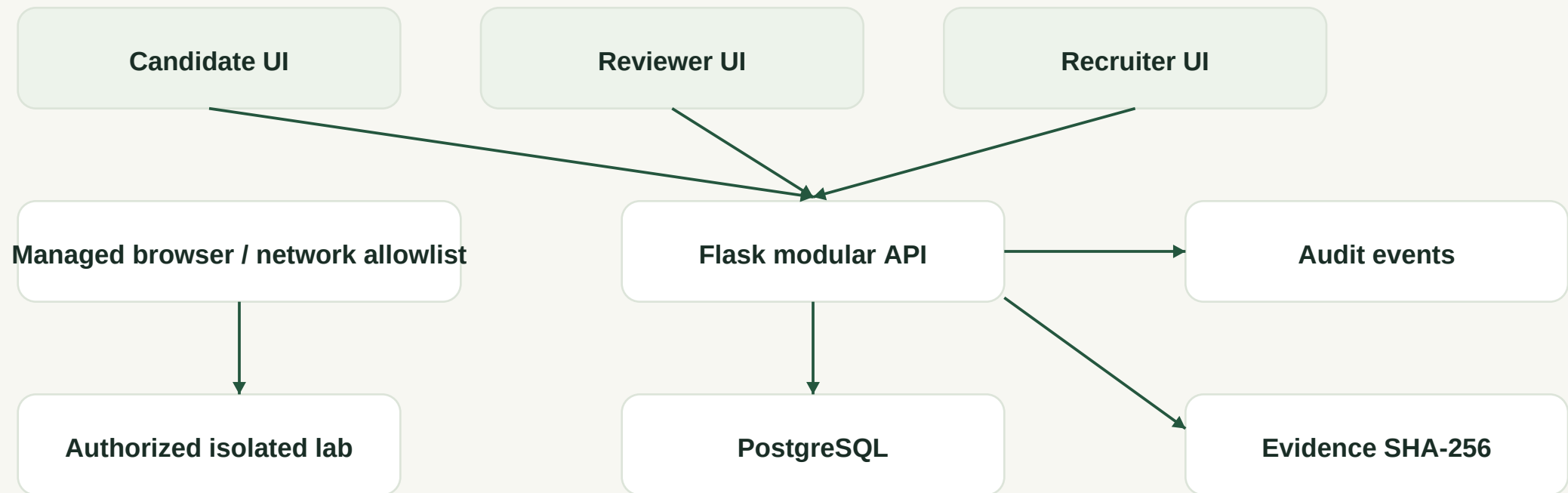
- 01 01 Candidate completes a section assessment and submits a structured finding.
- 02 02 API scores answers server-side and hashes submitted evidence.
- 03 03 Reviewer reproduces, questions and records a decision.
- 04 04 Verified records inform capability and proof-confidence signals.
- 05 05 Recruiter inspects sanitized proof and can invite the candidate.

## PRODUCT CAPTURE NEEDED

No real product screenshots are present in the repository. Replace this marked panel with a capture of the running product before submission.

# ARCHITECTURE

React + TypeScript role workspaces → Flask modular API → PostgreSQL



API also records audit events and computes SHA-256 evidence hashes.

Candidate UI → authorized lab (current lab UI is synthetic; no isolated target is provisioned).

# CONSTRAINTS & HONEST STATUS

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Evidence integrity · SHA-256 hash and duplicate flag in API | PARTIAL

Jurisdiction · No isolated lab provisioned; do not test arbitrary targets | NOT BUILT

Priority · Human reviewer controls verification | HANDLED

Bad input · API validates assessment shape and finding fields | PARTIAL

Offline · Local SQLite fallback; no offline sync guarantee | PARTIAL

# THE KEY DECISION

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## CHOSEN

Deterministic server-side scoring + structured evidence + human reviewer authority

## REJECTED

LLM as final grader: faster feedback, but less repeatable and harder to defend on adversarial evidence.

## TRADE-OFF ACCEPTED

Human review takes longer and creates a reviewer-capacity bottleneck. The MVP favors explainable decisions.

# WHAT THE CODE PROVES

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## API flow

Assessment answer keys stay server-side; each section score is persisted.

## Review flow

Human reviewer decisions, defense responses and rubric values have API persistence.

## Discovery boundary

API sanitizes verified proof. Some candidate and recruiter screens still use synthetic data.

## REAL CAPTURE SLOTS

Add 2–3 genuine screenshots: candidate assessment, reviewer decision, recruiter sanitized proof. No screenshots are checked in.

# LIMITS & SCALE

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**50,000 candidates × 4 sections = 200,000 score submissions.**

- Over 20 minutes: ~167 requests/second average; deadline bursts are higher.
- First pressure point: synchronous score + separate audit commits; one configured worker; in-memory rate limit.
- First change: durable idempotent submission/outbox queue; shared Redis limits; then load-test.
- Other gaps: synthetic UI data; no isolated labs; no secure file-upload/redaction pipeline.

# NEXT & DISCLOSURE

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1 Connect role pages to authenticated persisted records.

2 Provision isolated, resettable labs with egress denied.

3 Add shared rate limiting, monitoring and peak-load tests.

AI disclosure: Claude/Gemini assisted development; no runtime LLM makes hiring decisions.

**Live MVP:** <https://proofforge-cyber.onrender.com>

Repository: [github.com/yogesh-37911/HM26-C3B5-Phase2](https://github.com/yogesh-37911/HM26-C3B5-Phase2)

Team members / colleges: [add names]